The first NERSC campaigns: stability, heating and equilibrium correlators

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(Dated: November 10, 2026)

Abstract

Three campaigns of the full stochastic scheme (Hamiltonian step plus the stochastic viscous step in the particle representation) were run on Perlmutter on 8–9 November 2026: a stability scan in the thermal velocity and λ (`first`, `first_det`), and equilibrium ensembles in boxes $L = 1.2$ and 2.4, once with the old code and a thermal start (`first`) and once with the fast code and the equilibrium start (`second`). This note says what was run, what each figure shows and why it was made, and what we learned. In short: (i) the runs heat slowly, as δv^4 and faster than linearly in time, and $\lambda = 1$ blows up between $\delta v = 0.078$ and 0.14; (ii) the transverse momentum correlator is not diffusive but a damped oscillation, transverse sound of a weakly rigid particle arrangement; (iii) the fast code and the equilibrium start reproduce the old results.

I. WHAT WAS RUN

a. The scheme. Every run is a uniform two-dimensional conformal fluid ($c_s^2 = 1/3$, $\eta/s = 0.2$) in a periodic box on a hexagonal lattice with $h = 0.05$ and $h/a = 2$. A time step is the leapfrog of the ideal SPH Hamiltonian H between stochastic viscous steps (stress and residual moves, checkerboard sweeps, `stochrep.tex`), with the gradient-corrected change of representation and the star vote. The parameter λ of the residual move is $\lambda = 1$ (detailed balance: every mode, including the lattice modes, fluctuates) or $\lambda = \infty$ (the residual is deterministic: the lattice modes are damped without noise and stay cold). The entropy density S_0 sets the size of the fluctuations: the entropy per particle is $s_a = S_0 V_a$ and the thermal velocity per component is

$$\delta v = \Big(\frac{T}{w V_a}\Big)^{1/2} = s_a^{-1/2}, \tag{1}$$

S_0	s_a	δv
10^6	541	0.043
3×10^5	162	0.078
10^5	54	0.14

The time step is $\Delta t = 8.66 \times 10^{-3}$ in all runs.

b. The campaigns. One Slurm array task per realization, 2 CPUs (one core) each, in the `shared` queue; job scripts, commits and commands are stored next to the output in `$SCRATCH/sph/<campaign>/`, and the record is `nersc/LOG.md`.

campaign	set	what varies	box	runs	to
`first`	ref	$S_0 = 10^6$, $\lambda = 1$ and $\lambda = \infty$	0.6	16	$t = 200$
	s3e5	$S_0 = 3 \times 10^5$, $\lambda = 1$	0.6	16	200
	s1e5	$S_0 = 10^5$, $\lambda = 1$	0.6	16	200
	s1e5_ha3	$S_0 = 10^5$, $\lambda = 1$, $h/a = 3$	0.6	16	200
	s1e5_hfix	$S_0 = 10^5$, $\lambda = 1$, energy fix	0.6	16	failed
`first_det`	s3e5_det, s1e5_det	$\lambda = \infty$, same seeds	0.6	16+16	200
`rigidity`	Sect. V	h/a, kernel, λ, no H step, no viscous step	1.2, 0.6	16 per set	200
`energystep`	Sect. VI	$H = \sum_a e_a$	1.2, 0.6	16+4	200
`first`	eqL1.2, eqL2.4	thermal start, old code	1.2, 2.4	32+32	400
`second`	eqL1.2, eqL2.4	equilibrium start, fast code	1.2, 2.4	32+32	400

The boxes 0.6, 1.2, 2.4 hold $N = 672$, 2688, 10560 particles. In `first` only 12 of the 32 eqL2.4 runs finished within the 8 h limit (the others were lost, since output was written only at the end); `second` writes a checkpoint every 20 minutes and runs about twice as fast per core (Sect. VIII).

c. The two drivers.

- `stochrep_hamiltonian.jl` (stability scan) starts from rest with all momenta zero, and records over time the kinetic energy $\sum_a |\boldsymbol{m}_a|^2$ divided by its equilibrium value $2(N-1)V_a w T$; the smoothed energy H (conserved by the leapfrog); the sum of the particle energies $\sum_a e_a$ (held fixed in a stochastic step); and the *local disorder*, the rms change of the distances between original nearest neighbours in units of a. A disorder above 1 means the lattice has melted.

- `stochrep_equilibrium.jl` (ensembles) records the Fourier modes of the momentum density, $\boldsymbol{M}_{\boldsymbol{k}}(t)$, split into the longitudinal and transverse parts M_L, M_T, for all $kh \leq 1.5$, every 2 (L = 1.2) or 4 (L = 2.4) steps. From them

$$C_{T,L}(k,\tau) = \frac{\langle M_{T,L}(\boldsymbol{k}, t+\tau) M^*_{T,L}(\boldsymbol{k}, t)\rangle}{N V_a w T}, \tag{2}$$

  averaged over $t > 2$, over the vectors $\boldsymbol{k}$ of a shell $|\boldsymbol{k}| = k$, and over the realizations. In equilibrium $C(k,0) = 1$. Hydrodynamics predicts

$$C_T(k,\tau) = e^{-\nu k^2 \tau}, \qquad C_L(k,\tau) = \cos(c_s k\tau)\, e^{-\Gamma_s k^2 \tau/2}, \qquad \nu = \frac{\eta}{w} = \frac{\eta/s}{T}, \tag{3}$$

  with $\Gamma_s = \frac{4}{3}\nu$ (conformal) and here $\nu = 2.2 \times 10^{-3}$.

II. STABILITY SCAN

a. Purpose. The earlier runs (`fullruns.tex`) at $\delta v = 0.14$ melted the lattice and blew up after $t \approx 45$, while at $\delta v = 0.043$ they only heated slowly. The scan locates the limit in δv, and asks whether the noise in the lattice modes ($\lambda = 1$ against $\lambda = \infty$, same seeds) or the resolution ($h/a = 3$) is responsible.

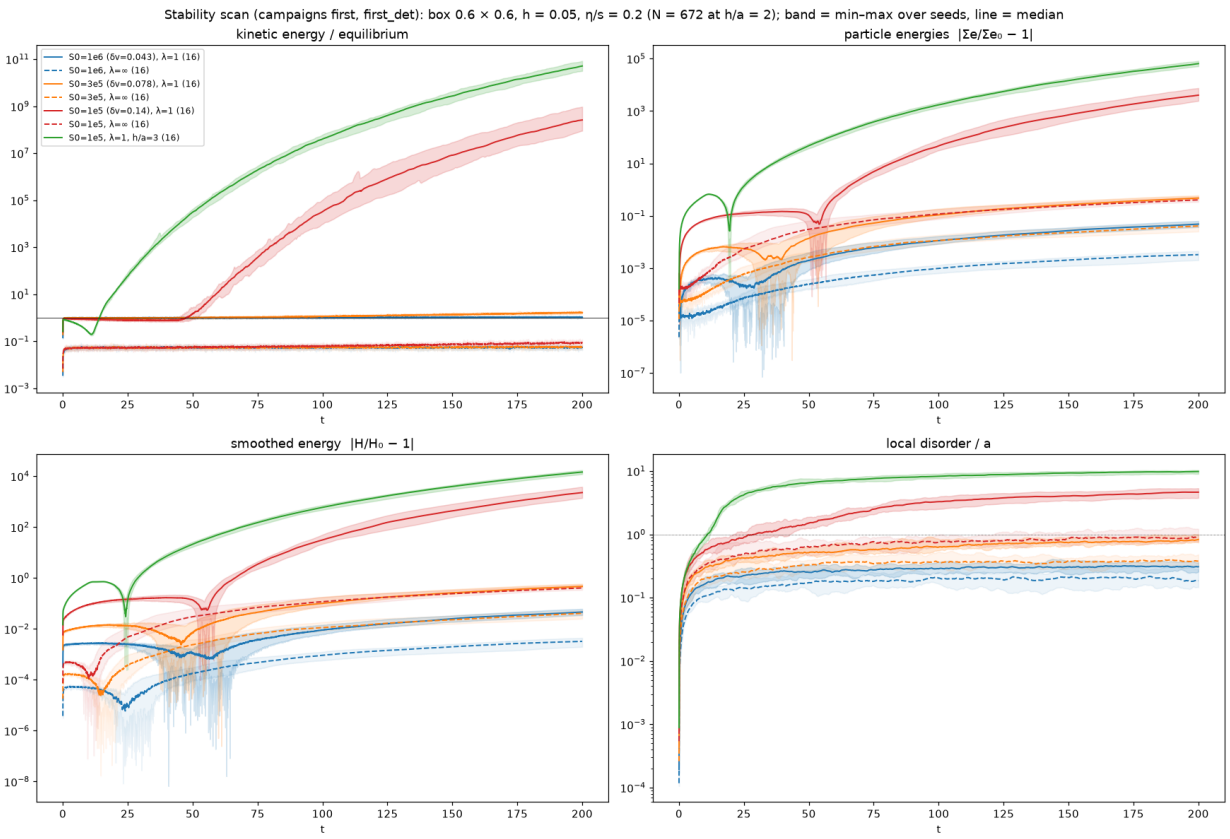


FIG. 1. Stability scan, box 0.6, $h/a = 2$ unless noted, 16 seeds per set. Line: median over the seeds; band: minimum to maximum. Colour: δv; solid $\lambda = 1$, dashed $\lambda = \infty$; green: $h/a = 3$. Top left: kinetic energy over its equilibrium value (the $\lambda = \infty$ runs sit at ≈ 0.05, because only the hydrodynamic modes are thermalized). Top right and bottom left: drift of the particle energies and of the smoothed energy, in magnitude (the dips are sign changes). Bottom right: local disorder; the dotted line marks melting.

b. What the figures show.

set ($t = 200$, median)	$H/H_0 - 1$	KE/eq	disorder	melted	blew up
$\delta v = 0.043$, $\lambda = 1$	0.046	1.06	$0.31a$	0/16	0
$\delta v = 0.043$, $\lambda = \infty$	0.0032	0.05	$0.19a$	0/16	0
$\delta v = 0.078$, $\lambda = 1$	0.47	1.66	$0.83a$	1/16	0
$\delta v = 0.078$, $\lambda = \infty$	0.041	0.06	$0.38a$	0/16	0
$\delta v = 0.14$, $\lambda = 1$	2×10^3	3×10^8	$4.7a$	16/16 ($t \approx 28$)	16
$\delta v = 0.14$, $\lambda = \infty$	0.40	0.09	$0.93a$	8/16 ($t \approx 147$)	0
$\delta v = 0.14$, $\lambda = 1$, $h/a = 3$	1.5×10^4	5×10^{10}	$9.9a$	16/16 ($t \approx 11$)	16

1. *The limit.* With $\lambda = 1$ the scheme is stable at $\delta v = 0.078$ (one seed just melting at $t = 198$) and blows up at $\delta v = 0.14$: the instability sets in between $s_a = 54$ and 162. Once the disorder reaches about a the kinetic energy grows exponentially.

2. *The lattice modes.* $\lambda = \infty$ never blows up, even where its lattice disorders to about a.

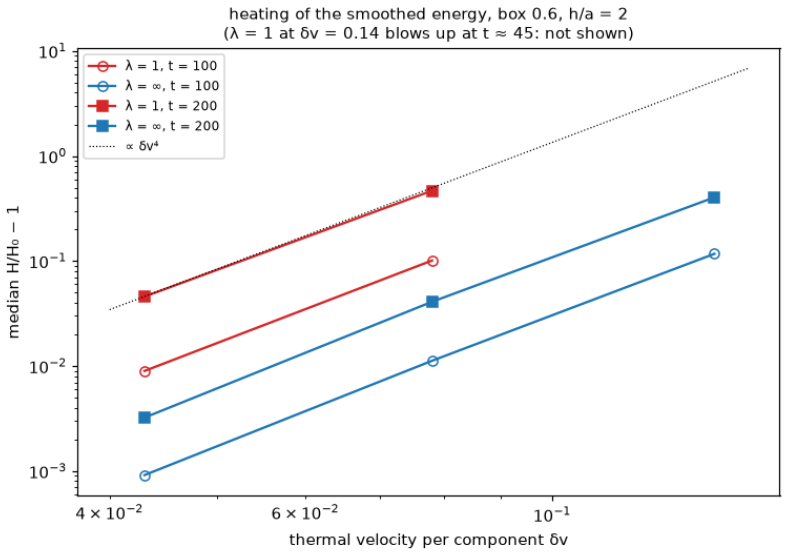


FIG. 2. The heating of Fig. 1: median $H/H_0 - 1$ at $t = 100$ (open) and $t = 200$ (filled) against δv, log–log. Dotted: $\propto \delta v^4$.

The runaway therefore needs the noise in the lattice modes.

3. *The heating* (Fig. 2) grows as δv^4 for both λ, and $\lambda = 1$ heats 10–14 times faster than $\lambda = \infty$ at the same δv. Between $t = 100$ and 200 the drift grows by a factor 4–5: it accelerates, roughly as t^2, rather than growing linearly.

4. $h/a = 3$ blows up sooner. At the same S_0 it has more, smaller particles, so a larger δv (0.21): not a test at fixed δv.

5. *The energy fix* (`SR_HFIX=1`): the runs failed in the bond weights once the lattice had melted, both before and after a fallback was added; the fix does not prevent the blow-up.

III. EQUILIBRIUM CORRELATORS

a. Purpose. Test eq. (3): do the stochastic step and the Hamiltonian step together reproduce the equilibrium fluctuations ($C(k, 0) = 1$) and the hydrodynamic decay of the shear and sound modes? The boxes are large enough to reach $kh = 0.26$ ($L = 1.2$) and 0.13 ($L = 2.4$). All at $\delta v = 0.043$, $\lambda = 1$, $t = 400$.

b. Fig. 3: first look.

- The sound modes oscillate at the right frequency, $c_s k$, and decay at about the predicted rate.

- The statics are 1.04–1.10 for $kh \gtrsim 0.5$: 5–10% high, about what the slow heating does to wT over the run. At the smallest k, $C_T(k, 0) \approx 0.9$.

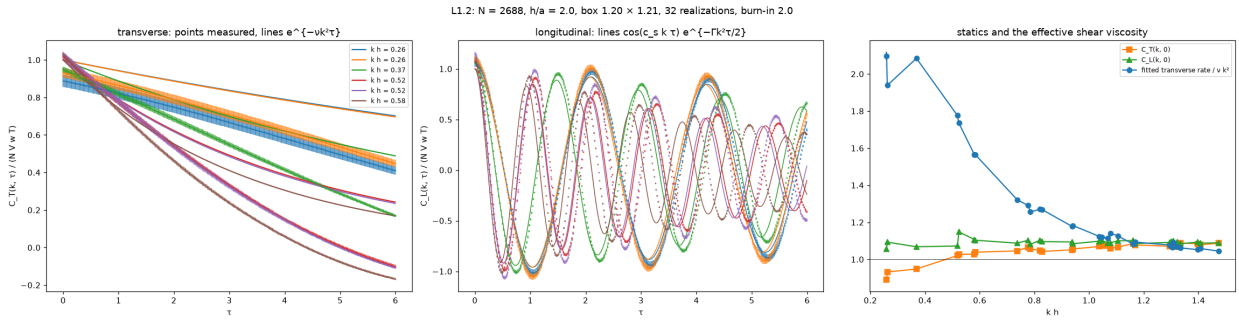


FIG. 3. Campaign `first`, $L = 1.2$, 32 realizations, lags to $\tau = 6$. Left: $C_T(k,\tau)$ (points with error bars) and $e^{-\nu k^2 \tau}$ (lines) in the six smallest shells. Middle: $C_L(k,\tau)$ and the sound prediction of eq. (3). Right: the statics $C_T(k,0)$, $C_L(k,0)$, and the transverse decay rate fitted at short lags as an exponential, in units of νk^2.

- At short lags C_T falls faster than $e^{-\nu k^2 \tau}$ at small k, by a factor 2 in the rate at $kh = 0.26$. The longer lags of the next figures show why.

c. Fig. 4: the transverse correlator is an oscillation. To long lags C_T crosses zero, reaches about -0.4, and comes back: it oscillates, at a frequency proportional to k. This is transverse sound, which a fluid does not have and an elastic solid does. The particles never melt at $\delta v = 0.043$ (disorder $0.3a$), and the SPH energy of a sheared lattice is not exactly that of the unsheared one, so the arrangement responds as a weakly rigid solid. The simplest model is a viscous fluid with a small shear modulus G (Kelvin–Voigt): the transverse displacement u obeys

$$\ddot{u} = -\nu k^2 \dot{u} - c_t^2 k^2 u, \qquad c_t^2 = \frac{G}{w}, \tag{4}$$

and the momentum correlator is the velocity correlator of a damped oscillator,

$$C_T(k,\tau) = A\, e^{-\gamma\tau}\Big[\cos\omega\tau - \frac{\gamma}{\omega}\sin\omega\tau\Big], \qquad \gamma = \frac{\nu k^2}{2}, \qquad \omega^2 = c_t^2 k^2 - \gamma^2. \tag{5}$$

Its initial slope, $-2\gamma = -\nu k^2$, is the hydrodynamic one: at short lags the oscillator and $e^{-\nu k^2 \tau}$ agree, and they part once $c_t k \tau \sim 1$. We fit A, γ and ω in each shell. The fits (dashed) lie on the data within the errors over the whole range, with $A = 1.0$–1.07.

d. Fig. 5: what the fits give.

- *A constant transverse sound speed*, $c_t = 0.022 \pm 0.001$ for $kh = 0.13$–0.74, the same in both boxes and both campaigns: a linear dispersion, as for an elastic solid. Then

$$\frac{G}{p} = \frac{w\, c_t^2}{p} = 4c_t^2 \approx 2 \times 10^{-3} \qquad (w = 4p), \tag{6}$$

a shear modulus three orders of magnitude below the pressure. It is still large enough to dominate the shear mode at $kh \lesssim 0.5$, where $c_t k > \nu k^2/2$, i.e. for

$$k < \frac{2c_t}{\nu} \approx 20, \qquad kh < 1. \tag{7}$$

- *The damping is close to the viscous one at* $kh \geq 0.5$: $2\gamma/\nu k^2 = 1.1$–1.2. At smaller k it rises, to 1.3–1.5 at $kh = 0.26$ and 1.9–2.1 at $kh = 0.13$ (32 runs at $L = 2.4$). The fit recovers the damping of synthetic data to $\pm 7\%$ with 8 runs (Sect. IV), so this excess is real. The extra rate,

$$2\gamma - \nu k^2 \approx 0.015,\ 0.023,\ 0.036 \qquad \text{at} \qquad kh = 0.13,\ 0.26,\ 0.52, \tag{8}$$

  grows roughly as $k^{2/3}$, not as k^2: it is not a viscosity. Its origin is open (a slow relaxation of the arrangement, or the rise of T over the run).

- The two campaigns agree shell by shell: the oscillation is not an artifact of the start or of the code changes.

So the earlier statement that the effective shear viscosity is twice the input one at small k (Fig. 3, right) is wrong: it was an exponential fitted to the onset of the oscillation. The viscosity is about right; what is wrong is the rigidity.

A. The two campaigns side by side

a. Purpose. `second` repeats the ensembles of `first` with the fast code and the equilibrium start. It checks the code changes on long runs and shows what the start changes.

b. What the figures show.

	runs	$H/H_0 - 1$ at $t = 400$	KE/eq at $t = 400$	$C_T(k,0)$, smallest shell
`first`, $L = 1.2$	32	0.201 ± 0.010	1.28	0.89
`second`, $L = 1.2$	32	0.219 ± 0.009	1.31	0.96
`first`, $L = 2.4$	12	0.199 ± 0.005	1.28	0.69 ($kh = 0.13$)
`second`, $L = 2.4$	32	0.221 ± 0.004	1.31	0.75 ($kh = 0.13$)

(the $\pm$ is the spread over the runs).

- *The correlators agree.* $C_T(k,\tau)$ is the same oscillation in both campaigns, and the statics agree within their scatter. The fast code and the new start change nothing measurable in equilibrium.

- *The start.* From the thermal start H stays flat for $t \lesssim 30$ and then heats (the particle energies start without fluctuations, `eqstart.tex`, which may be why); from the equilibrium start the heating begins at once. By $t = 400$ the two differ by 0.02 in $H/H_0 - 1$, about two spreads.

- *The heating is the same in every box.* $H/H_0 - 1 \approx 0.2$ and KE/eq ≈ 1.3 at $t = 400$, growing faster than linearly in time, as in the small box at the same δv (Sect. II). It is a property per particle, not of the box size. Over the run T rises by about 5%, which is why the statics at large k sit at 1.05–1.10.

- *The statics at small* k are low: $C_T(k,0) = 0.89$–0.96 at $kh = 0.26$ and 0.65–0.75 at $kh = 0.13$, and $C_L(k,0) \approx 0.93$ at $kh = 0.13$. A possible cause is the rigidity: the runs start with the particles on the lattice, so the elastic energy of the long transverse modes starts at zero, and is filled only at the slow rate νk^2 ($1/\nu k^2 \approx 70$ at $kh = 0.13$), drawing on the kinetic energy. The displacements are recorded in the runs now going, which will test this.

IV. IS THE OSCILLATION REAL?

Three checks of the analysis (scripts on the branch `rigidity`).

1. *An independent implementation.* `examples/check_correlators.py` reads the raw $M_T(\boldsymbol{k}, t)$ with h5py and computes C_T by direct sums over t (no FFT, no shared code). For `second`, $L = 1.2$, at $kh = 0.26$ it gives $C_T = 0.9575$, 0.5635, -0.3695 at $\tau = 0$, 5, 20, the same to four digits as the Julia analysis.

2. *Synthetic data.* `test/testcorrelators.jl` generates transverse modes with a known shear modulus and viscosity,

$$\dot{u} = v, \qquad dv = (-\nu k^2 v - c_t^2 k^2 u)\, dt + \sqrt{2\nu k^2}\, dW, \tag{9}$$

   sampled as in the runs (same k shells, recording interval and length, 8 runs), and puts them through the same correlator and fit.

input	kh	c_t fitted	$2\gamma/\nu k^2$ fitted
$c_t = 0.022$	0.26, 0.37, 0.52	0.0220, 0.0216, 0.0219	0.98, 1.02, 1.02
$c_t = 0$ (fluid)	0.26, 0.37, 0.52	0.000, 0.000, 0.003	0.93, 1.07, 0.97

   The analysis recovers both, and does not produce an oscillation from a fluid.

3. *Physical controls*, running (Sect. IX 0 a): positions frozen (no Hamiltonian step), where C_T must be $e^{-\nu k^2 \tau}$; $\lambda = \infty$; and c_t measured a second way, from the recorded displacements $u_T(\boldsymbol{k}, t)$, by

$$c_t^2 = \frac{\langle |M_T|^2 \rangle}{(wV)^2 k^2 \langle |u_T|^2 \rangle}. \tag{10}$$

V. THE RIGIDITY CAMPAIGN

a. Purpose. Find what sets the transverse sound speed c_t of Sect. III: the smoothing length in units of the spacing, the kernel, the noise in the lattice modes, or the viscous step. Campaign `rigidity` (jobs 61045626, 61045885, 61046892; `nersc/campaign_rigidity*.txt`): $L = 1.2$, $t = 200$, the equilibrium start, $\delta v = 0.043$, 16 realizations per set, and the particle displacements $\boldsymbol{u}_a = \boldsymbol{x}_a - \boldsymbol{x}_a^0$ recorded besides $\boldsymbol{M}_{\boldsymbol{k}}$. The sets: C2 at $h/a = 2$, 2.5, 3; Wendland C4 at $h/a = 2$, 2.5 and C6 at $h/a = 2$; $\lambda = \infty$; the positions frozen (no Hamiltonian step); and the ideal step alone (no viscous step, thermal start, $L = 0.6$, $t = 50$). Analysis `examples/rigidity_analysis.jl`.

Two measures of c_t in each shell. The *fit*: $C_T(k, \tau)/C_T(k, 0)$ fitted to the damped oscillator of Sect. III. The *displacements*: an elastic mode has $\langle |\dot{u}_T|^2 \rangle = c_t^2 k^2 \langle |u_T|^2 \rangle$, and $\boldsymbol{M}_{\boldsymbol{k}} \simeq wV_a \dot{\boldsymbol{u}}_{\boldsymbol{k}}$, so

$$c_t^2 = \frac{\langle |M_T(k)|^2 \rangle}{(wV_a)^2 k^2 \langle |u_T(k)|^2 \rangle}, \tag{11}$$

which goes to zero in a fluid, where $\langle |u_T|^2 \rangle$ grows without bound.

set	c_t (fit)	c_t (displacements)	$2\gamma/\nu k^2$
C2, $h/a = 2$	0.022	0.017	1.26–1.63
C2, $h/a = 2.5$	0.015	0.012	1.06–1.54
C2, $h/a = 3$	0.012	0.008	1.04–1.62
C4, $h/a = 2$	0.023	0.017	1.34–2.17
C4, $h/a = 2.5$	0.016	0.011	1.29–1.59
C6, $h/a = 2$	0.026	0.019	1.25–1.90
C2, $h/a = 2$, $\lambda = \infty$	0.020	0.015	1.49–1.79
positions frozen	0.000–0.003	–	0.99–1.09

(c_t at $kh = 0.37$; the damping over $kh = 0.26$–0.52.)

1. The viscous step alone is a fluid: with the positions frozen $C_T = e^{-\nu k^2 \tau}$ with the input viscosity. The rigidity and the extra damping come with the Hamiltonian step.

2. c_t falls with h/a, roughly as $(h/a)^{-1}$, and does not fall with the order of the kernel: C4 and C6 are as stiff as C2 or stiffer.

3. The static phonon test of `rigidity.tex` (the $T = 0$ lattice) does not predict these numbers: there the C4 lattice at $h/a = 2$ is unstable to shear and C6 is half as stiff as C2. The rigidity at temperature T is not the harmonic lattice's.

4. The displacement estimate is 25–35% below the fit everywhere; not understood.

b. The ideal step alone (Fig. 9) runs away at every h/a and kernel: KE/eq reaches 40–90 by $t = 50$ while H is conserved, the kinetic energy taken from the internal energy.

VI. THE ENERGY REPRESENTATION

a. What changes. The ideal step of `energystep.tex` (`SPH_HAMILTONIAN=energy`, now on `main`; the default is still `smoothed`). The Hamiltonian is the sum of the particle energies,

$$H = \sum_a e_a\big(s_a, \boldsymbol{m}_a, V_a\big), \qquad V_a = \frac{s_a}{S_a}, \qquad S_a = \sum_b s_b W_{ab}, \tag{12}$$

each evaluated with the particle's own momentum density $\boldsymbol{m}_a/V_a$. Hamilton's equations then move each particle with its own velocity, $\dot{\boldsymbol{x}}_a = \boldsymbol{v}_a$, instead of a smoothed one, and give a pure pressure force. The stochastic step is unchanged except that the gradient correction of the change of representation is off (it is zero for this H). Everything else (C2, $h/a = 2$, $\eta/s = 0.2$, $\delta v = 0.043$, $\lambda = 1$, Δt) is as in the `smoothed` runs it is compared with.

b. 1. Equilibration and heating (debug job 61055621; box 0.6, $t = 200$, 4 realizations per curve), Fig. 10. Purpose: does it run, does it equilibrate, does it heat? Left, the full scheme from a cold start (all momenta zero); right, the ideal step alone (no viscous step) from the thermal start.

1. Both reach KE/eq ≈ 1 within $t \approx 1$ from the cold start. `smoothed` then heats faster than linearly ($\Delta H/H = +5.7\times10^{-2}$ at $t = 200$, KE/eq drifting to 1.05–1.10); `:energy` stays at KE/eq = 0.99.

2. `:energy` instead loses energy at a constant rate, the same in every realization,

$$\frac{1}{H}\frac{dH}{dt} \simeq -3.4 \times 10^{-5}. \tag{13}$$

   Its source is being tested (campaign `energy2`: Δt, frozen positions).

3. The ideal step alone: `smoothed` conserves H but its kinetic energy runs away (KE/eq 60–95 by $t = 200$; Sect. V shows the same for every h/a and kernel). `:energy` settles at KE/eq ≈ 0.67, with H drifting by $+1.4 \times 10^{-6}$ per unit time.

c. 2. Transverse correlator (jobs 61067905, 61073581; $L = 1.2$, $t = 200$, 16 realizations, against the `smoothed` set C2, $h/a = 2$ of Sect. V), Fig. 11. Purpose: is the shear mode now the diffusive mode of eq. (3)?

1. `smoothed`: in every shell C_T crosses zero and swings to -0.2 to -0.4, the transverse sound of Sect. III.

2. `:energy`: no zero crossing in any shell; C_T lies on or just below $e^{-\nu k^2 \tau}$. Fitted to the damped oscillator, $c_t = 0.000$–0.002 (against 0.020–0.023) and $2\gamma/\nu k^2 = 1.03$–1.33 (against 1.26–1.63), Fig. 13.

3. $C_T(k, 0)$ is 0.93–1.04 (against 0.89–0.97); with errors of 0.02–0.07 both are within about 2σ of one, so the earlier deficit at small k is not significant here.

d. 3. Longitudinal correlator (same runs), Fig. 12. Purpose: does the new ideal step keep the sound mode? Both carry sound at c_s. With `:energy` the oscillation stays in phase with the prediction to $\tau = 30$ and decays slightly faster than it; with `smoothed` it falls behind in phase and decays more slowly. $C_L(k, 0) = 0.89$–1.06 for both.

e. 4. All observables together , Fig. 13: the run averages, the fit of C_T, and the transverse displacements, which with `:energy` grow far beyond the `smoothed` plateau (they are taken by minimum image in the box, so they wrap once the particles have moved far, which is why the curve turns down).

f. Cost. Per step `:energy` is slightly cheaper than `smoothed` (59 against 64 s for 4 realizations to $t = 10$ on 8 threads); the stochastic step dominates. (The `energystep` jobs ran 5 times slower than this because of a kernel-dispatch bug, since fixed; `nersc/LOG.md`.)

g. Why the rigidity goes away is not established. The static shear modulus of the lattice is the same for both Hamiltonians (at $\boldsymbol{m} = 0$ they are the same function of the positions), and the ideal step alone is still a solid (KE/eq ≈ 0.67, equipartition with the elastic energy). A possible reason: with $\dot{\boldsymbol{x}}_a = \boldsymbol{v}_a$ neighbouring particles can slip past one another under the noise, while the smoothed velocity moves them together and keeps the arrangement locked.

VII. RANDOM PARTICLE POSITIONS

If the rigidity comes from the hexagonal lattice, a random arrangement should not have it. Short tests (box 0.6, $t \leq 10$; positions uniform at random, the s_a rescaled so that the smoothed entropy is uniform, then the equilibrium start, which now handles unequal volumes):

run	t	KE/eq	$H/H_0 - 1$
full scheme	10	bursts to 150	+0.64, rising steadily
ideal SPH only	5	420	10^{-4} (conserved)
stochastic step only, positions frozen	5	0.65–0.68	10^{-4}
the same on the lattice	5	≈ 1.0	10^{-4}

At fixed particle entropies the SPH energy depends on the arrangement: the random one lies about $400T$ per particle above the lattice, released as motion by the ideal step and turned into heat by the viscous step. So the equilibrium of the particle positions at temperature T is near a minimum of this energy, a solid, and a random arrangement is not an equilibrium state at all. In the continuum the energy of a perfect fluid is invariant under volume-preserving relabelings of the fluid elements, which is why a fluid has no shear modulus; the SPH energy breaks this symmetry at the scale a, by an amount set by the kernel's error on the lattice. A fluid needs that breaking to be $\lesssim T$ per particle, or the particles to be relabeled (moved relative to the fluid with the fields held fixed).

VIII. CODE CHANGES BETWEEN THE CAMPAIGNS

a. Speed. The serial parts (smoothing, velocities, Hamiltonian force, thermodynamics, the bond weights) are threaded, the neighbour list is reused, and the conformal equation of state is inverted in closed form. Time per particle and step at $L = 2.4$:

threads	1	2	8
before	52 μs	42 μs	37 μs
after	27 μs	13 μs	5.5 μs

Validation before `second`: 32 against 32 realizations ($L = 0.6$, $t = 50$), old against new code; all 13 observables (KE/eq, $H/H_0 - 1$, the two acceptances, $C_{T,L}(k, 0)$ and $C_T(k, \tau = 5)$ in three shells) agree within 1.3σ.

b. Start. `second` starts from the equilibrium draw of `eqstart.tex` (energies and momenta at one global β^μ, totals shifted to the target), so no thermalization is needed; `first` started from `thermal_init!` (momenta at fixed energies).

c. Checkpoints. Every 20 minutes the state goes to `<run>.chk` and the record so far to the `.h5` (marked incomplete); resubmitting the task resumes it, bit for bit.

IX. WHAT WE LEARNED AND OPEN QUESTIONS

1. *Stability.* At $h/a = 2$ the scheme with $\lambda = 1$ is stable to $t = 200$ for $\delta v \leq 0.078$ ($s_a \geq 162$) and blows up at $\delta v = 0.14$. The blow-up needs the noise in the lattice modes: $\lambda = \infty$ never blows up.

2. *Heating.* All runs heat, at a rate $\propto \delta v^4$, 10–14 times faster with $\lambda = 1$ than with $\lambda = \infty$, and accelerating in time. At $\delta v = 0.043$ it is 20% in H by $t = 400$ in every box. Its source is not identified. A rate $\propto \delta v^4$ is second order in the fluctuations, as a time-step or representation error of the stochastic step would be.

3. *Transverse sound.* The particle arrangement has a shear modulus $G \approx 2 \times 10^{-3} p$ ($c_t = 0.022$). It turns the shear mode into a damped oscillation for $kh \lesssim 1$, which is the whole hydrodynamic range we can reach. With it removed from the fit, the shear viscosity comes out 10–20% above the input one at $kh \geq 0.5$; at smaller k there is an extra damping that does not scale as k^2.

4. *Code.* The fast code and the equilibrium start reproduce the old ensembles.

5. *Rigidity* (Sect. V). It comes with the `:smoothed` Hamiltonian step, falls as $(h/a)^{-1}$, not with the kernel order, and is gone with the energy representation (Sect. VI).

a. Next. The energy representation: understand its linear loss of H, check it at $h/a = 2.5$, 3 and in the $L = 2.4$ box ($kh = 0.13$), and redo the stability scan with it.

b. Open questions.

1. Where does the rigidity come from, and how does it scale? Candidates: the SPH energy of a sheared hexagonal lattice (a property of H alone), or the stochastic step pinning the arrangement. Tests: c_t against h/a and the kernel, in a run with the Hamiltonian step only (thermalized, no viscous step), and in a disordered (melted then cooled) arrangement. If it comes from H, a larger h/a should reduce it.

2. Where does the heating come from? Tests: its dependence on Δt (a step error falls as a power of Δt) and on η/s, with $\lambda = 1$ and $\lambda = \infty$.

3. Why are $C_T(k, 0)$, and less so $C_L(k, 0)$, below one at the smallest k, and where does the extra damping $\propto k^{2/3}$ come from?

4. Which particle-energy variance is the equilibrium one in the particle representation, the drawn one or 9/16 of it (`eqstart.tex`)?

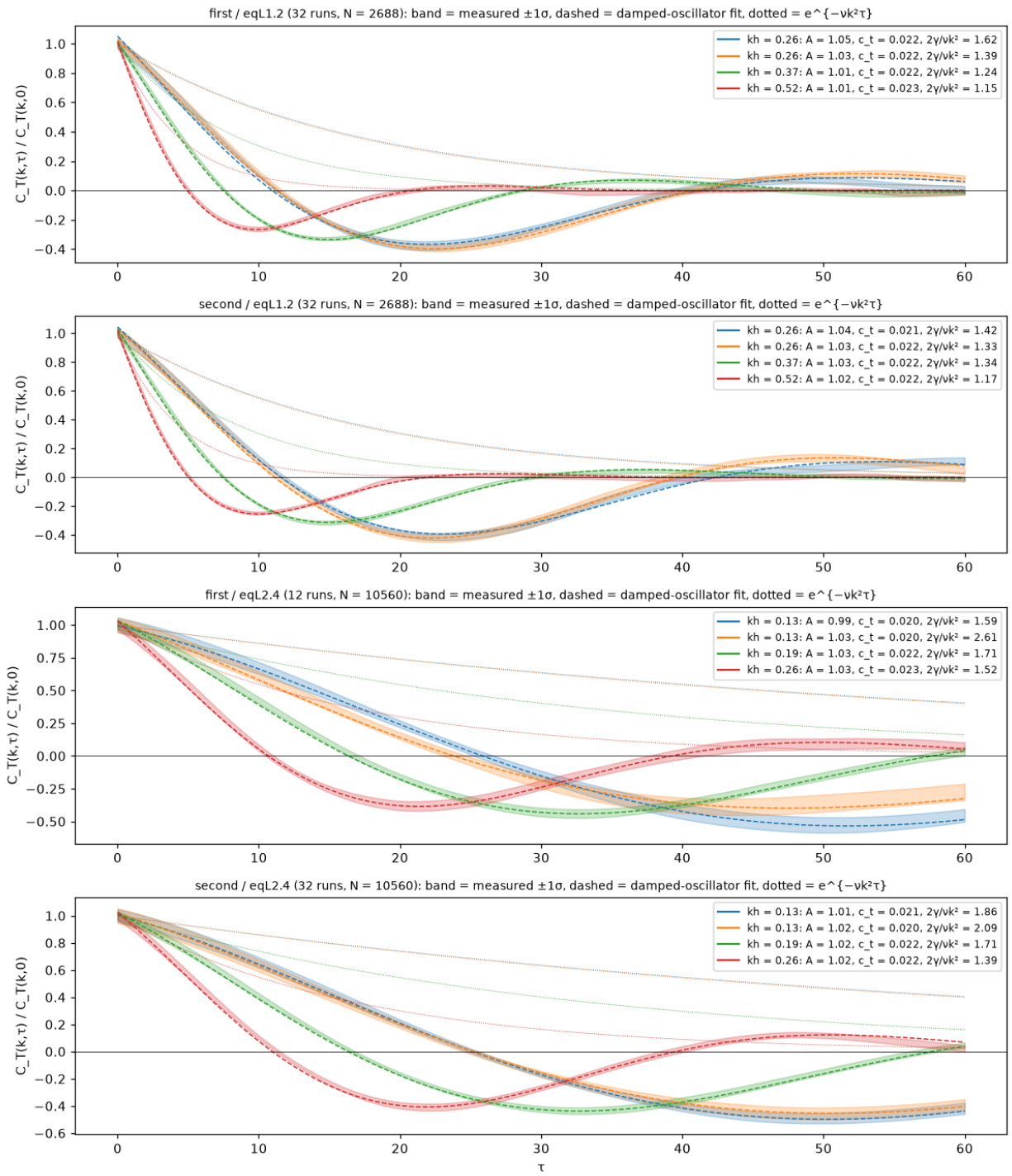


FIG. 4. $C_T(k,\tau)/C_T(k,0)$ to $\tau = 60$ in the four smallest shells. Band: measured mean $\pm 1\sigma$ over the realizations. Dashed: fit of the damped oscillator, eq. (5). Dotted: the hydrodynamic $e^{-\nu k^2 \tau}$. From top to bottom: `first` and `second` at $L = 1.2$ (32 runs each), `first` at $L = 2.4$ (the 12 runs that finished) and `second` at $L = 2.4$ (32 runs). The legend gives the fitted phase velocity ω/k and damping.

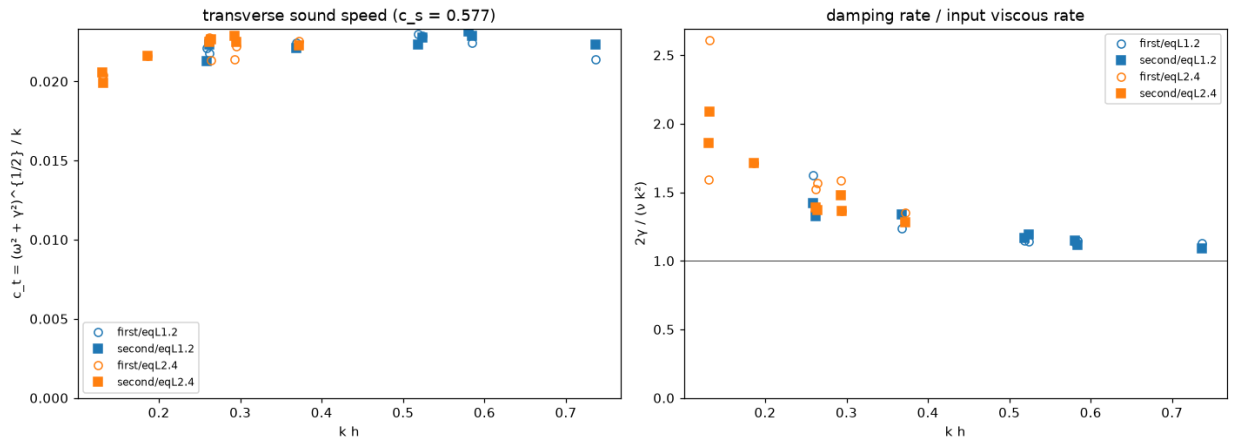


FIG. 5. The fits of Fig. 4 in the eight smallest shells. Left: the transverse sound speed $c_t = (\omega^2 + \gamma^2)^{1/2}/k$. Right: the damping rate in units of the viscous one, $2\gamma/\nu k^2$. Blue: $L = 1.2$; orange: $L = 2.4$; open: `first`, filled: `second`.

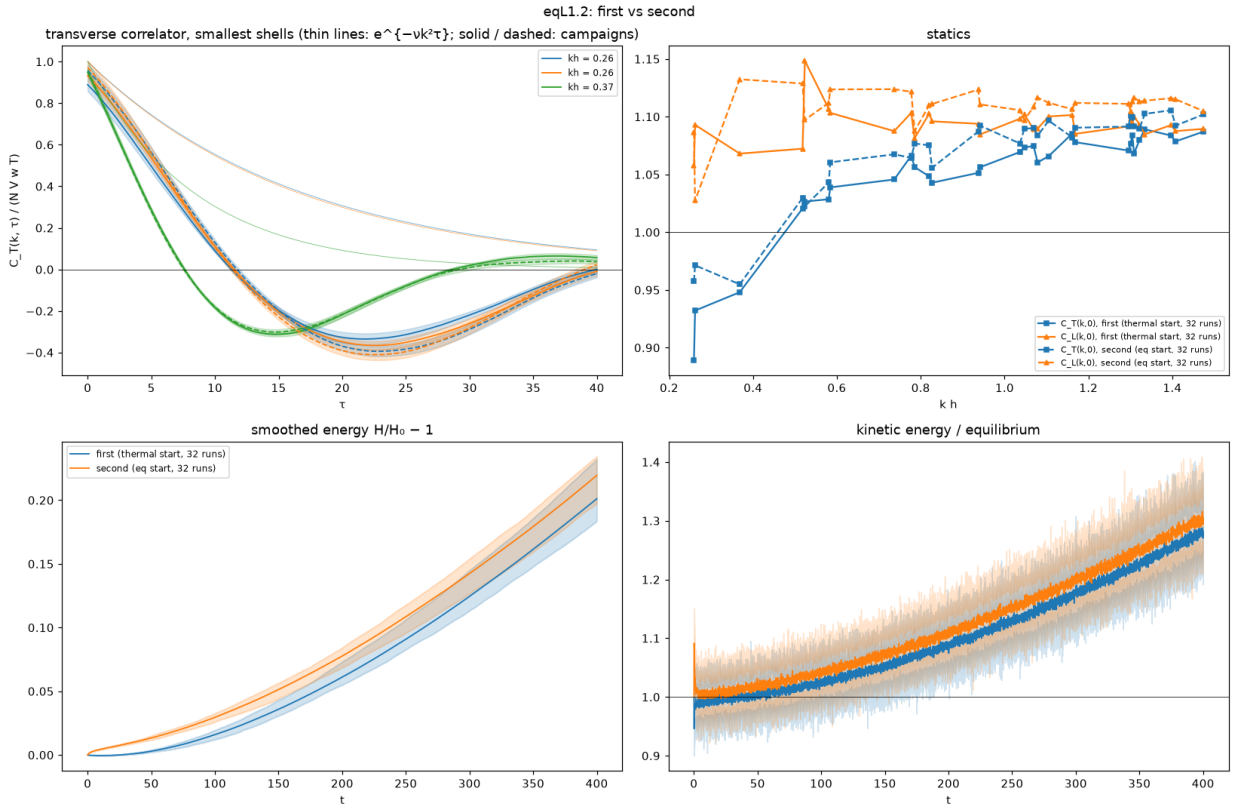


FIG. 6. $L = 1.2$, `first` (thermal start, old code, 32 runs) against `second` (equilibrium start, fast code, 32 runs). Top left: $C_T(k, \tau)$ in the three smallest shells, solid `first`, dashed `second`, thin lines $e^{-\nu k^2 \tau}$. Top right: the statics $C_T(k, 0)$ (blue) and $C_L(k, 0)$ (orange), normalized by the initial wT. Bottom: $H/H_0 - 1$ and the kinetic energy over its equilibrium value, mean over the runs (line) and minimum to maximum (band).

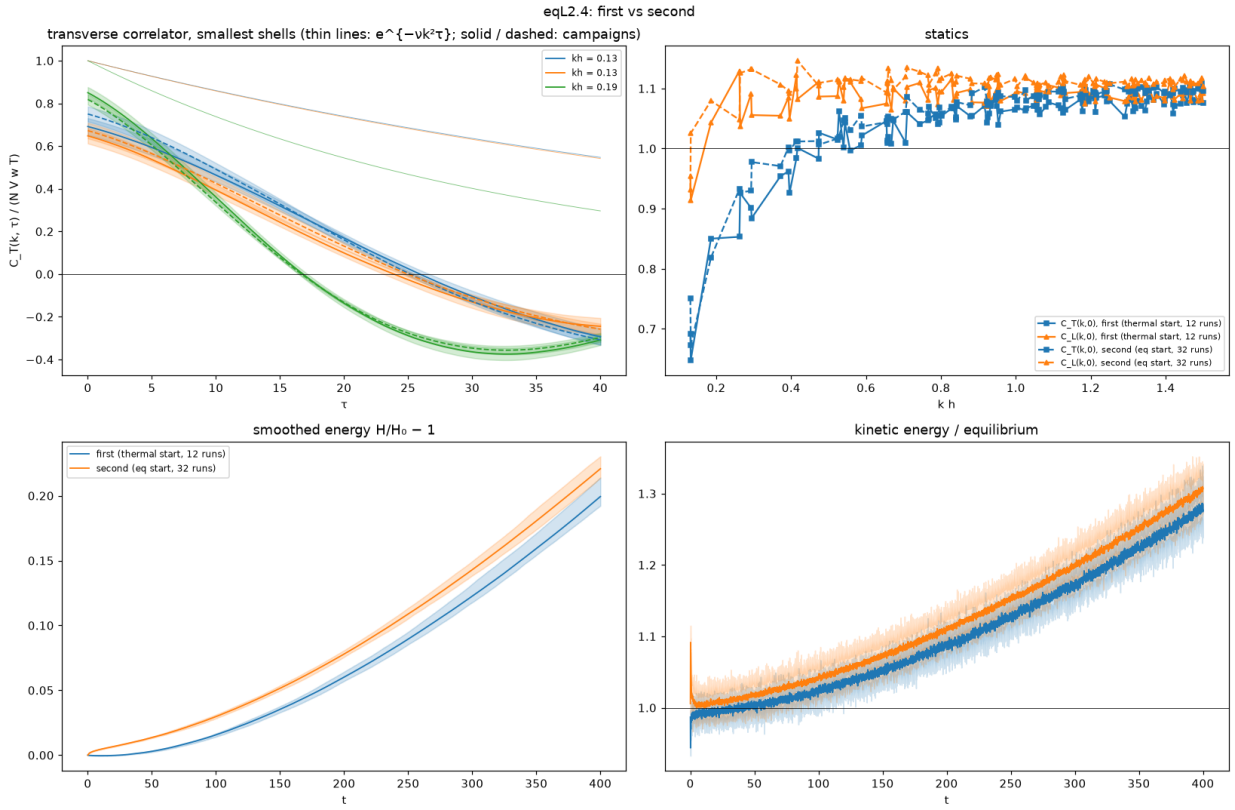


FIG. 7. As Fig. 6 for $L = 2.4$ ($N = 10560$, smallest shell $kh = 0.13$). `first`: the 12 runs that finished; `second`: all 32.

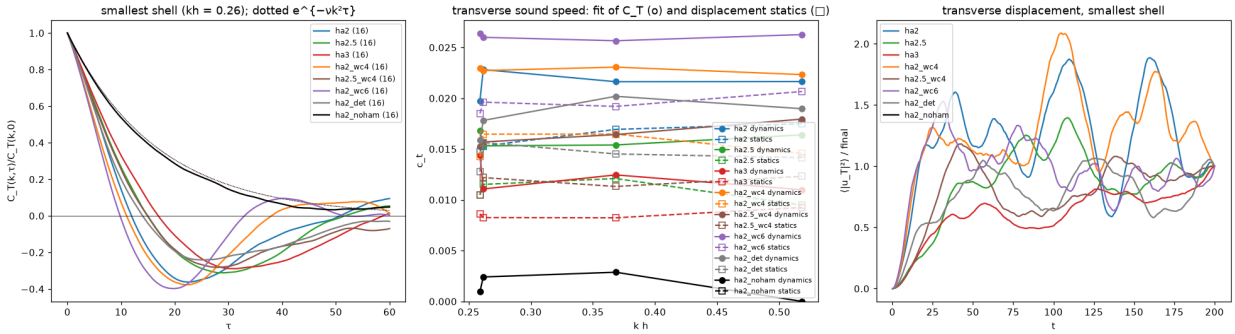


FIG. 8. Left: $C_T(k,\tau)/C_T(k,0)$ in the smallest shell for every set; dotted, the fluid $e^{-\nu k^2 \tau}$. Middle: c_t from the fit (circles) and from the displacements (squares) against kh. Right: $\langle |u_T(k,t)|^2 \rangle$ in the smallest shell, divided by its final value.

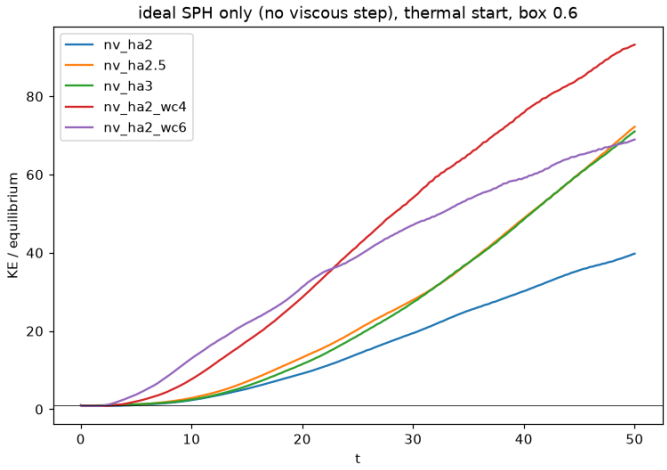


FIG. 9. The ideal step alone (`:smoothed`, no viscous step) from the thermal start, $L = 0.6$: KE/eq against t, 4 realizations per set.

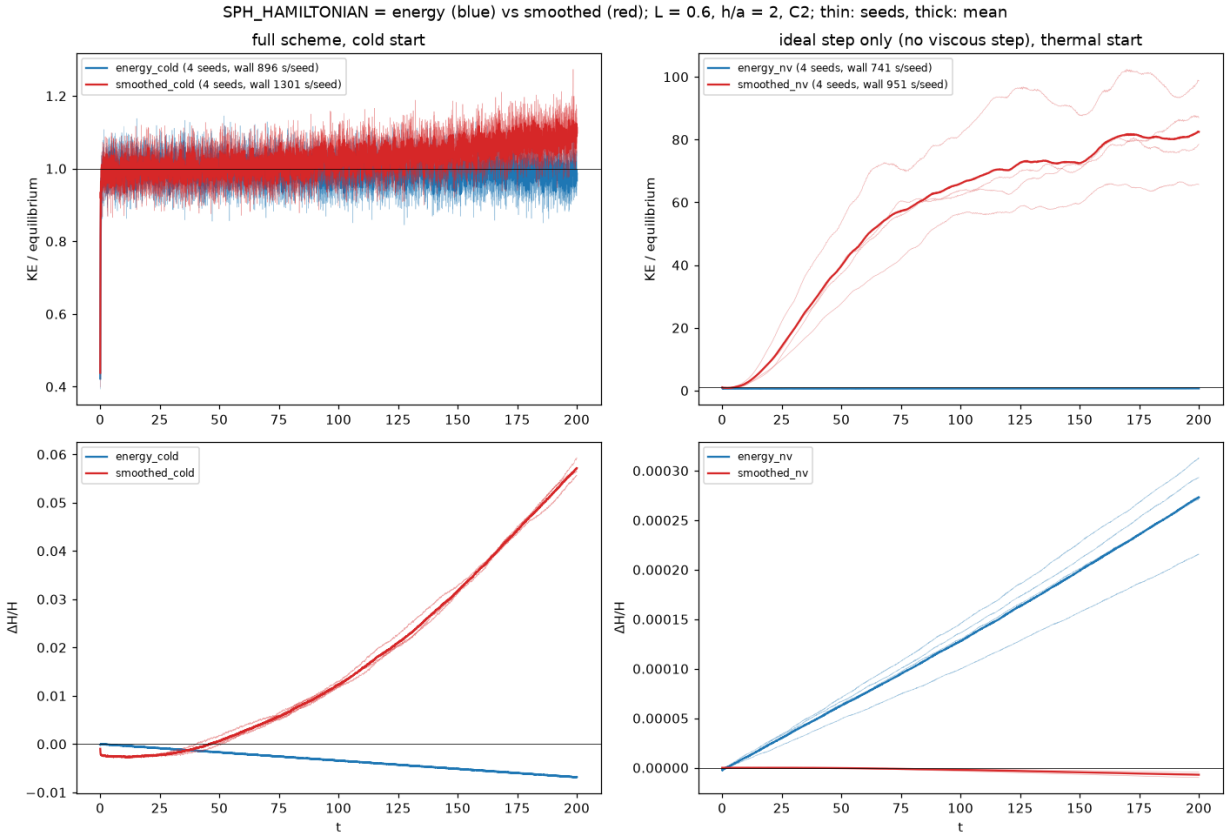


FIG. 10. `:energy` (blue) against `:smoothed` (red), box 0.6. Top: kinetic energy over its equilibrium value; bottom: $\Delta H/H$; thin lines single realizations, thick lines the mean. Left: full scheme, cold start. Right: ideal step only, thermal start. Script `examples/energystep_equilibration.jl`.

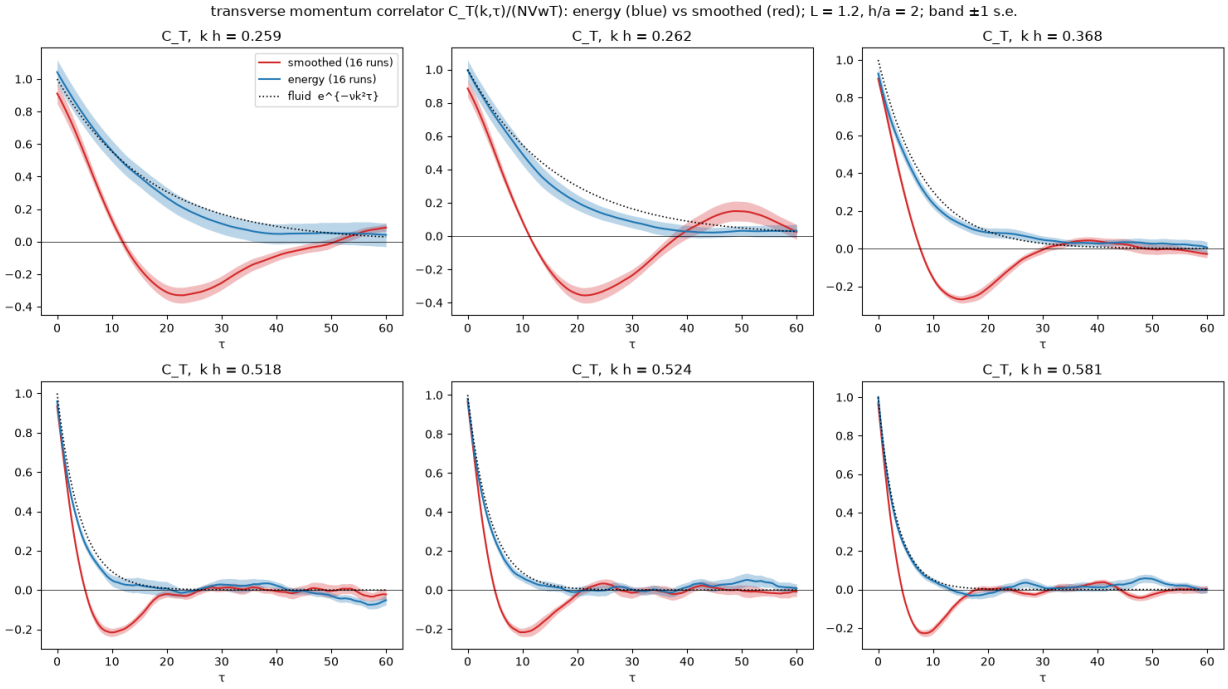


FIG. 11. $C_T(k,\tau)$, normalized by $NV_a wT$, in the six smallest shells: `:energy` (blue) and `smoothed` (red), bands ± 1 standard error over realizations; dotted, the fluid $e^{-\nu k^2 \tau}$ with the input ν (no fit). Script `examples/energystep_correlators.jl`.

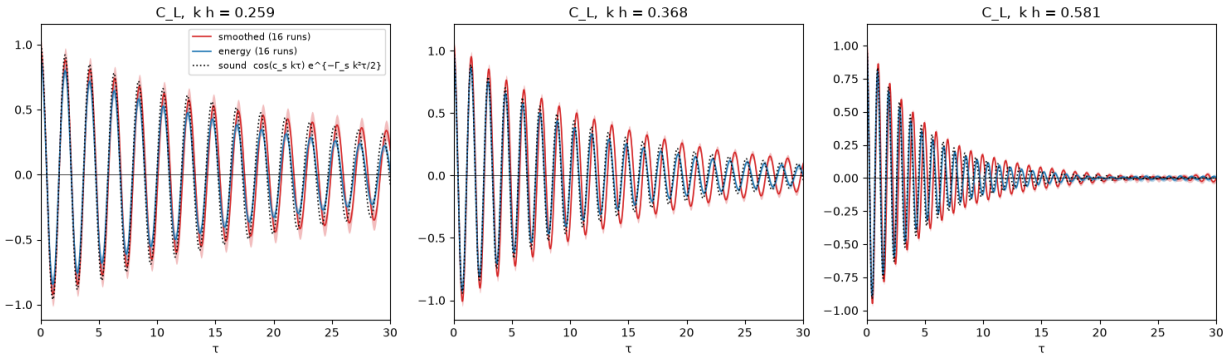


FIG. 12. $C_L(k,\tau)$ in three shells, `:energy` (blue) and `smoothed` (red), with the sound mode $\cos(c_s k\tau)\, e^{-\Gamma_s k^2 \tau/2}$ of eq. (3) (dotted, no fit).

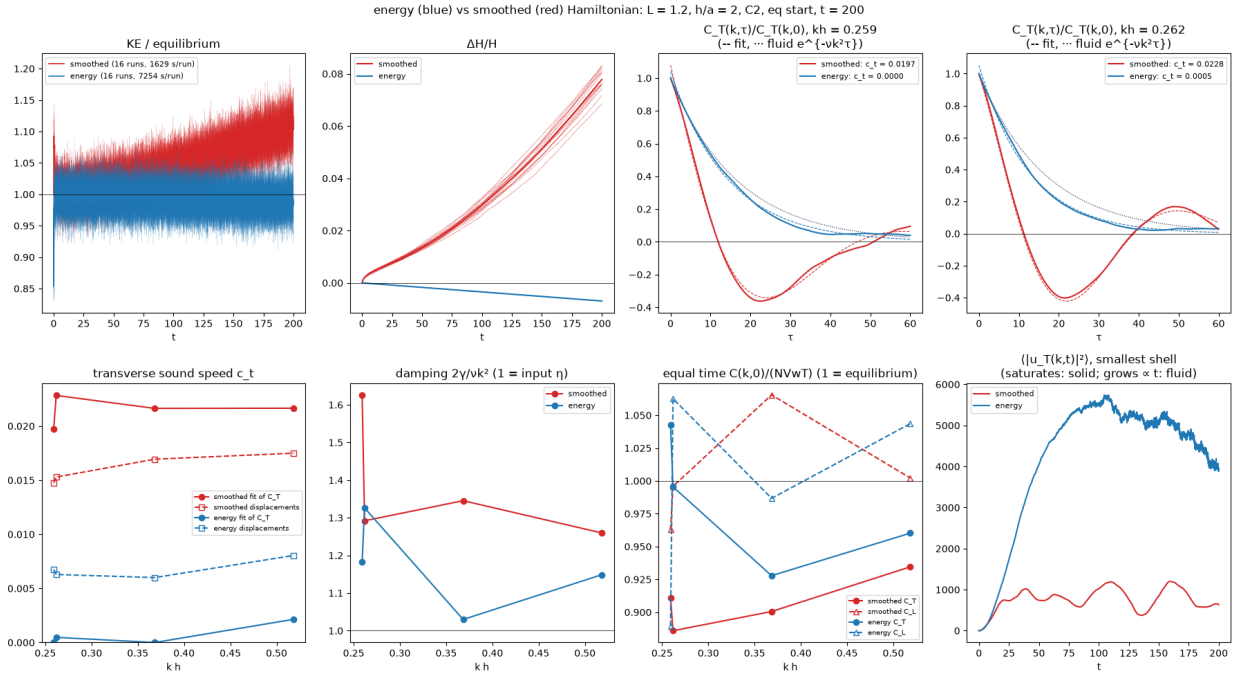


FIG. 13. `:energy` (blue) against `:smoothed` (red), $L = 1.2$, $h/a = 2$. Top: KE/eq, $\Delta H/H$, and $C_T(k,\tau)$ in the two smallest shells with the fit (dashed) and the fluid (dotted). Bottom: c_t from the fit (circles) and from the displacements (squares, meaningful only for a solid); the damping $2\gamma/\nu k^2$; the equal-time $C_{T,L}(k,0)$; and $\langle|u_T(k,t)|^2\rangle$ in the smallest shell. Script `examples/energystep_compare.jl`.